

Spotify & YouTube Music

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Step-by-Step Points for Data Cleaning and Processing

1. **Rearrange Data by Platform:**
 - Start by organizing your data first by Spotify-related information, then by YouTube-related information. This makes it easier to handle platform-specific data separately.
2. **Delete Unnecessary Columns:**
 - Identify and delete any columns that are not needed for your analysis. Removing these columns ensures that the data is more manageable and avoids confusion.
3. **Rename Columns:**
 - Change any 'Unnamed' columns to something more meaningful, such as 'Index'. This will help in accurately identifying and referencing the columns later.
4. **Remove Duplicate Values:**
 - Since the 'Index' column is a primary key in your dataset, make sure there are no duplicate values in this column to maintain data integrity.
5. **Separate Spotify Information:**
 - For the 'Spotify_Info' column, split the data by a delimiter (like a hyphen '-') to separate the album link and track ID into two distinct columns.
6. **Handle YouTube Information:**
 - For the 'YouTube_Info' column, split the text by a fixed number of characters (44) to separate the link to the video and the video title into two new columns.
 - Use the 'Extract' function to handle remaining hyphens, splitting the text before the delimiter (hyphen) from the end to extract relevant details.
7. **Clean Danceability Data:**
 - Replace any 'nan' values in the 'Danceability' column with 'null' values to avoid errors during data processing.
 - Convert the column data type from text to a decimal format for consistency and proper analysis.
8. **Clean Energy Data:**
 - Similar to the 'Danceability' column, replace 'nan' values in the 'Energy' column with 'null' values.
 - Change the data type from text to a decimal format to standardize the data for analysis.
9. **Remove Suffixes:**
 - For the 'Track' column, remove the suffix '_Track' by extracting text before the delimiter.
 - For the 'Artist' column, remove the prefix 'ARTIST_' by extracting text after the delimiter to keep the data clean and consistent.

Handling Missing Data and Rows

10. Fill Missing Values in 'Views' and 'Likes' Columns

- **Fill Missing Values with Averages:** Compute the average (mean) of the existing 'views' and 'likes' values and fill the 'null' entries with these averages.
- **Reason:** Filling missing values with the average provides a reasonable estimate that does not skew the overall dataset. It prevents loss of data due to missing values and maintains consistency in the dataset.

11. Remove Rows Without Video Links

- **Delete Rows with Missing Video Links:** Identify rows where the YouTube link is missing or empty, and remove them.
- **Reason:** Without a valid video link, the rows lack critical information necessary for analysis. Since these rows cannot contribute meaningful data, removing them ensures the dataset remains relevant and focused.

12. Clean the 'Album' Column

- **Remove Rows with 'NULL' in the Album Column:** Identify rows where the album column has a null value or is blank.
- Remove these rows from the dataset.
- **Reason:** The album information is essential for linking tracks to their respective albums on Spotify. Rows without this information are incomplete and do not provide value for analysis, so removing them ensures a clean and accurate dataset.

Overall Data Processing Workflow

- **Organize the data:** First, sort the data based on platforms to ensure clarity and efficiency.
- **Clean the data:** Remove unwanted columns and duplicates to maintain quality.
- **Transform the data:** Split and manipulate columns as needed to extract and structure the information properly.
- **Handle missing and erroneous data:** Replace 'nan' values and adjust data types for smoother processing.
- **Standardize the format:** Remove unnecessary prefixes and suffixes to keep the data uniform and easy to work with.
- **Ensure Data Relevance and Accuracy:** Confirm that all remaining data is relevant, complete, and accurate for meaningful analysis.

Key Points for Effective Data Cleaning:

- **Consistency:** Maintain a uniform approach to data handling to ensure the dataset remains consistent and reliable.

- **Accuracy:** Clean and format data accurately to avoid errors and misinterpretations.
- **Efficiency:** Remove irrelevant or redundant data to make the dataset more efficient and manageable.
- **Completeness:** Ensure all data is complete and meaningful for the intended analysis.